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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025
CS2201-1- CLOUD COMPUTING
(For CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each **Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **A major challenge in parallel programming is:**
A) Synchronization and communication overhead B) Reduced performance speed
C) Limited memory D) Sequential execution issues
2. **Data parallelism means:**
A) Applying the same operation to multiple data items simultaneously B) Executing different tasks on the same data
C) Using only distributed systems D) Breaking tasks into pipelines
3. **Service-Oriented Computing (SOC) emphasizes:**
A) Batch processing B) Tightly integrated monolithic applications
C) Loosely coupled, reusable services D) Only hardware virtualization
4. **The first step in transitioning from a classic data center to cloud is:**
A) Virtualization B) Consolidation
C) Automation D) IT as a Service
5. **Cloud computing is best described as:**
A) A personal computer service B) A new programming language
C) Delivery of computing services over the internet D) A type of hardware processor
6. **In the cloud reference model, IaaS provides:**
A) Only software applications B) Virtualized hardware resources like storage, servers, and networking
C) End-user tools only D) Development frameworks
7. **Cloud computing reduces CAPEX mainly by**
A) Replacing IT staff B) Increasing on-premises servers
C) Eliminating network costs D) Using pay-per-use OPEX model
8. **In virtualization, the software layer that allows multiple OS instances to share hardware resources is called:**
A) Guest OS B) Container
C) Kernel driver D) Virtual Machine Monitor
9. **Portability in virtualization allows:**
A) Migration of applications across OSes B) Movement of VMs across platforms
C) Removal of the hypervisor D) Virtual memory allocation
10. **Hardware-assisted virtualization uses extensions such as:**
A) Intel VT-x and AMD-V B) Hyper-Threading
C) NUMA D) RAID levels

11. The main goal of virtualization is:
 A) Reducing storage B) Limiting scalability
 C) Eliminating cloud computing D) Increasing hardware utilization
12. Which virtualization method has the least overhead?
 A) Full virtualization B) Hardware-assisted
 C) Paravirtualization D) Application virtualization
13. Which type of virtualization combines multiple physical storage devices into one logical storage pool?
 A) Network virtualization B) Compute virtualization
 C) Storage virtualization D) Application virtualization
14. VLANs are an example of:
 A) Storage virtualization B) Network virtualization
 C) Application virtualization D) Desktop virtualization
15. Azure offers which type of services?
 A) IaaS only B) SaaS only
 C) IaaS and SaaS only D) PaaS, IaaS, SaaS
16. Eucalyptus supports:
 A) PaaS only B) SaaS applications only
 C) IaaS private cloud deployment D) PaaS and SaaS
17. A unique feature of App Engine is:
 A) Manual scaling only B) Shared root access
 C) On-premise deployment D) Automatic scaling of applications
18. The Cloud Controller (CLC) in Eucalyptus:
 A) Manages overall resources and scheduling B) Provides VM lifecycle only
 C) Handles object replication only D) Monitors billing alone
19. Amazon EC2 provides:
 A) Storage services B) Virtual servers on demand
 C) Database replication D) Only monitoring
20. What does AWS IAM provide?
 A) Data encryption B) Monitoring
 C) Storage provisioning D) Identity and Access Management

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Outline SISD and MIMD hardware architectures for parallel processing with a neat diagram.	8	L*2	1	1
	b) Summarize the common architectural styles used in call and return category.	8	L2	1	1
2.	a) What is Service? Explain any four major characteristics of services.	8	L2	1	1
	b) Explain the concept of Infrastructure/Hardware as a Service (IaaS/HaaS) in the context of the cloud reference model.	8	L2	2	1
3.	a) Define community cloud. Explain different candidate sector for community cloud.	8	L2	1	1
	b) Outline the different types of cloud (Public, Private and Hybrid) with suitable examples.	8	L2	2	1
Unit – II					
4.	a) Explain VMware Full Virtualization reference model with the help of a neat diagram	8	L2	3	1

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| b) | Explain the concept of Desktop Virtualization and outline its benefits. | 8 | L2 | 3 | 1 |
| 5. a) | Explain Execution Virtualization with a diagram showing ISA, ABI, and API layers. Briefly describe privileged instructions and the difference between behavior-sensitive and control-sensitive instructions. | 8 | L2 | 3 | 1 |
| b) | Explain VMware workstation architecture with a neat diagram. | 8 | L2 | 3 | 1 |
| 6. a) | Summarize the importance of security monitoring in cloud computing and how it helps in risk mitigation. | 8 | L2 | 4 | 1 |
| b) | Explain the Cloud Security Alliance (CSA) approach to cloud security and its significance. | 8 | L2 | 4 | 1 |

Unit – III

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| 7. a) | Explain three different types of AWS EC2 instances and their typical use cases. | 8 | L2 | 5 | 1 |
| b) | Explain the Microsoft Azure compute service core concept. | 8 | L2 | 5 | 1 |
| 8. a) | Explain the components of Eucalyptus and their roles in private cloud architecture. | 8 | L2 | 5 | 1 |
| b) | Explain Google App Engine platform architecture with a neat diagram. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
